

KYLE NERO A. SANTOS

San Mateo Rizal, Philippines
kylesantos0411@gmail.com
Contact No: 09194850794
LinkedIn: <https://www.linkedin.com/in/kyle-nero-santos-14287b349/>
Portfolio: kylesantos0411.github.io/portfolio/



PROFESSIONAL SUMMARY

Electronics Engineering graduate specializing in embedded systems, computer vision, and software development. Experienced in designing hardware-software solutions using ESP32 and YOLO-based object detection, alongside developing cross-platform applications and enterprise workflow automation.

EDUCATION

Bachelor of Science in Electronics Engineering June 2022 – July 2026
Marikina Polytechnic College
Capstone: Development of an Automated Optical Detector for Microplastics in Rock Salt for Retail and Domestic Use
Achievements: 1st Placer in Engineering Capstone Project Exhibit (2026), Participated in Research Innovation Congress (2026)

INTERNSHIP

First Option Electronic System Phils., Inc. June 2025 - Aug 2025
Pasig City Metro Manila - Philippines

- Managed end-to-end surveillance operations from designing comprehensive CCTV layouts to overseeing control room personnel, enhancing overall site security and operational efficiency.

TECHNICAL SKILLS

Programming: Python, OpenCV, YOLOv11, C/C++, Arduino/ESP32 firmware logic, TypeScript, React Native, SQL
ITSM Automation: ServiceNow Basic Fundamentals, ServiceNow Catalog (Including and not limited to Creation of Catalog Item, Basic Flow Designer, And System Administration/Navigation)
AI and Computer Vision: model training, AI Automation, object detection, dataset annotation
API and Data Handling: API concepts, Supabase, integrating AI/API-backed features, Database Management, JSON
Tools and Simulation: VS Code, Antigravity IDE, Claude Code, Arduino IDE, Git/GitHub, Excel, MS Office

ENGINEERING EDUCATIONAL PROJECT EXPERIENCE

Capstone Project | MICROTECT: Automated Microplastic Detector April 2026

- Led the end-to-end system development of a Raspberry Pi 5 optical detection prototype, eliminating the time and cost limits of manual rock salt microscopy.
- Designed the complete software architecture, implementing a YOLOv11 and OpenCV detection workflow that achieved 0.826 precision, 0.733 recall, and 0.814 mAP@50.
- Developed an end-to-end operator interface, automating the generation of microplastic contamination and particle count reports.
- Executed controlled multi-trial testing across diverse rock salt samples, validating prototype performance under varied conditions.

Software Applications Project | HerCare and KKB Snap Product and Frontend Development

- Developed interactive mobile product flows for a women's health companion app (HerCare) using React, TypeScript, and Vite, delivering a seamless, user-focused interface.
- Developed a cross-platform receipt-splitting application using TypeScript, React Native, Expo, Supabase PostgreSQL, Realtime, and Mindee OCR, enabling receipt scanning, live item claiming, proportional bill calculation, and payment QR handoff.

CERTIFICATIONS

ServiceNow Administration Fundamentals In Progress

- Pursuing comprehensive platform administration training, preparing for the official Certified System Administrator (CSA) credential.
- Completed foundational system administrator microlearning modules, establishing a practical baseline for enterprise workflow management.

AFFILIATIONS

- Institute of Electronics Engineers of the Philippines, Student Member